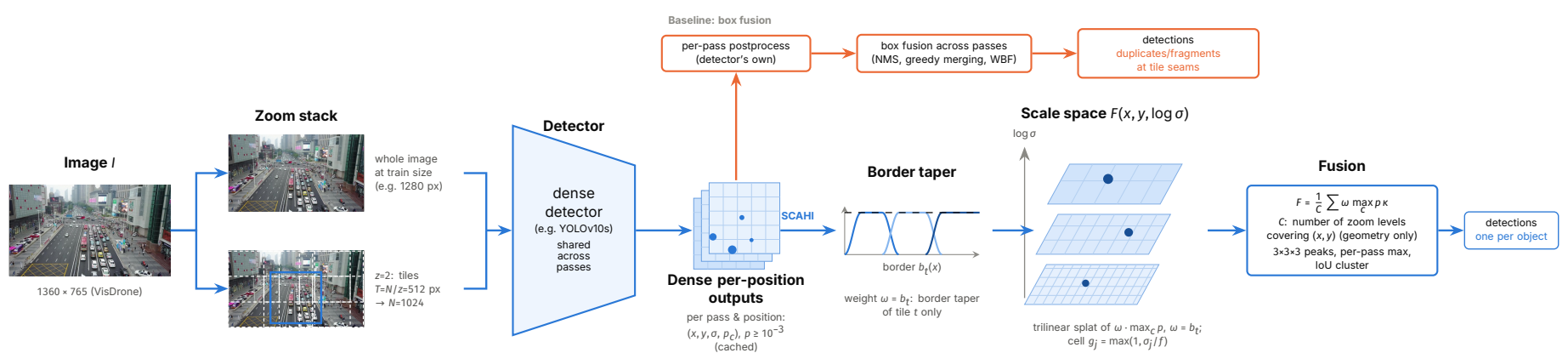

Baseline: box fusion
per-pass postprocess (detector's own)
box fusion across passes (NMS, greedy merging, WBF)
detections
duplicates/fragments at tile seams
Image I
1360 × 765 (VisDrone)
Zoom stack
whole image at train size (e.g. 1280 px)
z=2: tiles T=N/z=512 px → N=1024
Detector
dense detector
(e.g. YOLOv10s)
shared across passes
SCAHI
Dense per-position outputs
per pass & position:
(x, y, σ, p_c), p ≥ 10^−3
(cached)
Border taper
border b_t(x)
weight ω = b_t: border taper of tile t only
Scale space F(x, y, log σ)
log σ
trilinear splat of ω · max_c p, ω = b_t;
cell g_j = max(1, σ_j / f)
Fusion
F = 1/C Σ ω max_c p κ
C: number of zoom levels covering (x, y) (geometry only)
3×3×3 peaks, per-pass max, IoU cluster
detections
one per object